

Markov Processes

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September 15th, 2026

1 Finite-state Markov Chains

Until now, we have studied dynamical systems in which the evolution of the state is deterministic. Given the current state x_t , a deterministic law of motion determines exactly what the state will be next period such that

$$x_{t+1} = h(x_t)$$

Many economic environments, however, contain uncertainty. A worker may become employed or unemployed, a firm's productivity may increase or decrease, a borrower may or may not default, or the aggregate economy may move between expansions and recessions. In such settings, knowing the current state does not generally determine the next state with certainty. Instead, it determines the probabilities of the different states that may occur next.

Markov chains provide one of the simplest ways to model such stochastic dynamics. Rather than asking "What is the next state?", we ask "Given the current state, what are the probabilities of the possible next states?" This requires us to keep track not only of states themselves, but also of probability distributions over states.

1.1 Markov Processes

Suppose the dynamical system has a finite metric space $S = \{x_1, x_2, \dots, x_N\}$ and at each period t , the state of the system is represented by a random variable $X_t \in S$. This state space S contains all possible values the system can take. For example, if X_t represents the state of the economy, we might have

$$S = \{recession, boom\}$$

In principle, the probability of tomorrow's state could depend on the entire history $\{X_0, \dots, X_t\}$. Keeping track of the whole history quickly becomes cumbersome. Thus, a useful simplification can be done by assuming that the current state contains all the information from the past that is rele-

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vant for predicting the next state. This is called the **Markov property**. Using this, we can define a Markov chain as follows:

Definition 1. A discrete stochastic process X_t is called a **Markov chain** if

$$\mathbb{P}(X_{t+1}|X_t = x_t, \dots, X_0 = x_0) = \mathbb{P}(X_{t+1}|X_t = x_t)$$

That is, the probability of the random variable depends only on the most recent realizations. Thus, the past history does not matter.

The Markov assumption, where only the current state matters for predicting the next state, is less restrictive than it may initially appear. Suppose, for instance, that the evolution of X_{t+1} depends on both X_t and X_{t-1} . By defining $\hat{X}_t = (X_t, X_{t-1}) \in S \times S$, the process can be represented as a first-order Markov process in the enlarged state space. More generally, dependence on any finite number of past states can be incorporated into the definition of the current state.

Since the state X_t is a random variable, uncertainty about its realization can be summarized by the probabilities

$$P(X_t = x_1), \dots, P(X_t = x_N)$$

The collection of these probabilities forms a probability distribution over the state space.

Definition 2. Let $\mathcal{P}(S)$ denote the set of all probability distributions on S . A **probability distribution** $\phi \in \mathcal{P}(S)$ is a function $\phi : S \rightarrow [0, 1]$ that has the following properties:

1. $\phi(x) \geq 0 \forall x \in S$
2. $\sum_{x \in S} \phi(x) = 1$
3. $\phi(x)$ corresponds to the probability of x in some scenario.

Remark 1. An equivalent way to represent a distribution is as a vector $\phi = (\phi(x_1), \dots, \phi(x_N)) \in \mathbb{R}^N$ where $\phi_i = \phi(x_i)$. In this representation, the set of all distributions on S corresponds to the $(N - 1)$ -dimensional simplex $\Delta^{N-1} \subset \mathbb{R}^N$ where ϕ has the following properties:

1. $\phi_i \geq 0 \forall i$
2. $\sum_{i=1}^N \phi(x_i) = 1$

The simplex has dimension $N - 1$ because the condition $\sum_{i=1}^N \phi_i = 1$ imposes one linear restriction. For example, when $N = 3$, any distribution can be written as $\phi = (\phi_1, \phi_2, 1 - \phi_1 - \phi_2)$.

A probability distribution describes uncertainty about the state at a particular date. To describe how the system evolves over time, we also need to specify the probabilities of moving from one state to another.

By the Markov property, conditional on $X_t = x$, the distribution of X_{t+1} depends only on the realization of the current state x . Thus, for each $x \in S$, we need to specify the probabilities

$P(X_{t+1} = y | X_t = x)$ for every possible next state $y \in S$. For a fixed x , these probabilities form a probability distribution over next period's states. Collecting these conditional distributions for every $x \in S$ gives the transition rule of the Markov chain.

Definition 3. A **stochastic kernel** p is a function $p : S \times S \rightarrow [0, 1]$ such that

1. $p(x, y) \geq 0$ for all $(x, y) \in S \times S$
2. $\sum_{y \in S} p(x, y) = 1$ for all $x \in S$, i.e. the sum of each row is 1.

Remark 2. The finite state space stochastic kernels can be expressed as matrices such that

$$P = \begin{bmatrix} p(x_1, x_1) & \cdots & p(x_1, x_N) \\ \vdots & \ddots & \vdots \\ p(x_N, x_1) & \cdots & p(x_N, x_N) \end{bmatrix}$$

which is referred as a **transition matrix**. Using the conditional probability notation above, the transition matrix implies

$$P = \begin{bmatrix} \mathbb{P}(X_{t+1} = x_1 | X_t = x_1) & \cdots & \mathbb{P}(X_{t+1} = x_N | X_t = x_1) \\ \vdots & \ddots & \vdots \\ \mathbb{P}(X_{t+1} = x_1 | X_t = x_N) & \cdots & \mathbb{P}(X_{t+1} = x_N | X_t = x_N) \end{bmatrix}$$

The Markov property by itself does not require the transition probabilities to remain constant over time. In what follows, we focus on **time-homogeneous** Markov chains, for which the transition probabilities depend on the current and next states but not explicitly on time.

Definition 4. A Markov chain is **time homogeneous** if

$$\mathbb{P}(X_{t+1} = y | X_t = x) = \mathbb{P}(X_{s+1} = y | X_s = x)$$

for every $s, t \geq 0$ and every $x, y \in S$.

For a time-homogeneous Markov chain, we can therefore define

$$p(x, y) = \mathbb{P}(X_{t+1} = y | X_t = x)$$

where the right-hand side does not depend on t . Thus, $p(x, y)$ is the one period probability of moving from state x to state y .

Finally, to fully describe the stochastic evolution of the Markov chain, we also need to specify where the process starts. Since the initial state X_0 may itself be uncertain, this is described by an initial probability distribution.

Proposition 1. The stochastic property of a time-homogeneous Markov chain is completely determined by (1) its stochastic kernel p and (2) initial distribution $\psi_0 \in \mathcal{P}(S)$.

1.2 Marginal Distributions

Recall that the transition kernel describes how the system moves from one state to another, while the initial distribution describes our uncertainty about the state at $t = 0$. Together, these objects determine the stochastic evolution of the Markov chain.

We now ask “what is the probability of being in each state at a given future date t ?” This is described by the marginal distribution of X_t . For example, if the economy is known to be in a recession at $t = 0$, then the initial distribution assigns probability one to the recession state. The marginal distribution at some future state t describes the probability that the economy is in a recession or a boom at that date.

Definition 5. Let $\{X_t\}$ be a Markov— (p, ψ_0) , i.e. a Markov chain with transition probability p and initial distribution ψ_0 . For every $t \in \mathbb{N}$, the **marginal distribution** of X_t is the distribution $\psi_t \in \mathcal{P}(S)$ defined by

$$\psi_t(x) = \mathbb{P}(X_t = x) \quad \forall x \in S.$$

The marginal distribution is useful for forecasting because it describes the probability of the system being in each state at a given date. Once ψ_t is known, we can also compute expected outcomes at date t .

In practice, marginal distributions can be obtained either approximately through simulation or analytically using the law of total probability.

1.2.1 Simulation Method

The simulation method approximates the marginal distribution by simulating N independent trajectories of the Markov chain over T periods, given an initial state. At each period t , the proportion of simulated trajectories that are in state x provides an estimate of $\psi_t(x)$. As the number of simulated trajectories N becomes large, these empirical frequencies converge to the corresponding probabilities by the *law of large numbers*.

One limitation of this method is that it introduces a simulation error and may be computationally costly when a large number of trajectories is required.

1.2.2 Law of Total Probability

For a finite-state Markov chain, the marginal distribution can also be computed directly using the *law of total probability*. For any $y \in S$,

$$\begin{aligned} \mathbb{P}(X_{t+1} = y) &= \sum_{x \in S} \mathbb{P}(X_{t+1} = y | X_t = x) \cdot \mathbb{P}(X_t = x) \\ &\Rightarrow \psi_{t+1}(y) = \sum_{x \in S} p(x, y) \psi_t(x). \end{aligned}$$

since $\psi_t(x) = \mathbb{P}(X_t = x)$. Representing the marginal distribution as a row vector,

$$\psi_t = [\psi_t(x_1) \quad \cdots \quad \psi_t(x_N)],$$

the law of motion for the marginal distribution can be written compactly as

$$\psi_{t+1} = \psi_t \cdot P.$$

Explicitly,

$$[\psi_{t+1}(x_1) \quad \cdots \quad \psi_{t+1}(x_N)] = [\psi_t(x_1) \quad \cdots \quad \psi_t(x_N)] \begin{bmatrix} p(x_1, x_1) & \cdots & p(x_1, x_N) \\ \vdots & \ddots & \vdots \\ p(x_N, x_1) & \cdots & p(x_N, x_N) \end{bmatrix}.$$

Iterating the law of motion gives

$$\psi_t = \psi_0 \cdot P^t.$$

Thus, although the realized state X_t evolves randomly, its marginal distribution evolves according to the deterministic law of motion

$$\psi_{t+1} = \psi_t \cdot P.$$

In this sense, a finite-state Markov chain induces a deterministic dynamical system on the space of probability distributions.

1.3 Leftward and Rightward Markov Operators

1.3.1 A Linear Algebra Interpretation

Before introducing the Markov operator, it is useful to recall how matrix multiplication can be written as summation. Let

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

First, let $x \in \mathbb{R}^n$ be a column vector. Then the i -th element of Ax is

$$(Ax)_i = \sum_{j=1}^n a_{ij}x_j.$$

Thus, multiplying a matrix by a column vector fixes a row i and sum across the columns.

Now let $x \in \mathbb{R}^m$ be represented as a row vector. Then the j -th element of xA is

$$(xA)_j = \sum_{i=1}^m x_i a_{ij}.$$

Thus, multiplying a row vector by a matrix fixes a column j and sums across the rows.

This is exactly the structure that appears in a finite-state Markov chain. Since $\psi_{t+1} = \psi_t P$, the probability assigned to state x_j next period is

$$\psi_{t+1}(x_j) = \sum_{i=1}^N \psi_t(x_i) p(x_i, x_j).$$

Hence, the summation formula

$$\psi_{t+1}(y) = \sum_{x \in S} \psi_t(x) p(x, y)$$

is simply the component-wise representation of the matrix product $\psi_t P$, i.e. the matrix multiplication applied to a row vector.

1.3.2 Leftward Operation of M

We have just seen that in the finite-state case,

$$\psi_{t+1} = \psi_t \cdot P$$

is equivalent to

$$\psi_{t+1}(y) = \sum_{x \in S} \psi_t(x) p(x, y).$$

which describes how a distribution evolves when the transition kernel is represented by a matrix. The Markov operator expresses this same operation without relying on the matrix representation.

Definition 6. Given a kernel p , the **Markov operator** corresponding to p is the map $M : \mathcal{P}(S) \rightarrow \mathcal{P}(S)$ where

$$\psi M(y) = \sum_{x \in S} p(x, y) \cdot \psi(x) \forall y \in S$$

or equivalently, if X_t has the distribution ψ_t then the distribution of X_{t+1} is

$$\psi_{t+1} = \psi_t M$$

Remark 3. The operator M gives an abstract representation of the law of motion for distributions. In the finite-state case, if distributions are represented as row vectors, the same operation is represented by matrix multiplications:

- **Markov:** $\psi_{t+1} = \psi_t M$
- **Matrix:** $\psi'_{t+1} = \psi'_t P$

Hence, M and P represent the same transition rule in operator and matrix notation, respectively.

Repeating the Markov operation gives

$$\psi_{j+k} = \psi_j M^k$$

Thus,

$$\psi_j, \psi_j M, \psi_j M^2, \dots$$

forms a trajectory of marginal distributions generated by the Markov operator.

Remark 4. A special case arises when the state at some date is known with certainty, i.e. when $\psi_t = \delta_X$, implying

$$\mathbb{P}(X_t = x) = 1 \text{ and } \mathbb{P}(X_t \neq x) = 0$$

which is called a **degenerate distribution**.

The kernel $p(x, y)$ describes the probability of moving from x to y in one period. We can similarly define the probability of moving between states over multiple periods.

Definition 7. The t -th order kernel is

$$p^t(x, y) = \sum_{z \in S} p^{t-1}(x, z)p(z, y)$$

where $p^1 \equiv p$.

The interpretation is straightforward. To move from x to y in t periods, the process must first reach some intermediate state z after $t - 1$ periods and then move from z to y in the final period.

Exercise 1. Prove that p^t is a proper kernel, i.e. $p(x, y) \geq 0$ and $\sum_{y \in S} p(x, y) = 1$ for all $x \in S$.

Proof. We prove by mathematical induction. For $t = 1$, we have $p^1 = p(z, y)$. Thus, it is non-negative and each row adds up to one for all $x \in S$. Now suppose p^{t-1} is a stochastic kernel and consider p^t . First,

$$p^t(x, y) = \sum_{z \in S} p^{t-1}(x, z)p(z, y) \geq 0$$

since it is a product of two nonnegative terms. Now,

$$\begin{aligned} \sum_{y \in S} p^t(x, y) &= \sum_{y \in S} \sum_{z \in S} p^{t-1}(x, z)p(z, y) \\ &= \sum_{z \in S} p^{t-1}(x, z) \cdot \sum_{y \in S} p(z, y) \cdots \text{ since } |S| < \infty \\ &= \sum_{z \in S} p^{t-1}(x, z) \cdot 1 \cdots \text{ by case } t = 1 \\ &= 1 \end{aligned}$$

where the last equality hold by the assumption that p^{t-1} is a stochastic kernel. Therefore, p^t is a proper kernel. $\square$

Lemma 1. If M is a Markov operator defined by p on S , then M^t is the Markov operator defined by p^t such that

$$\psi M^t(y) = \sum_{x \in S} p^t(x, y)\psi(x)$$

1.3.3 Rightward Operation of M

So far, the Markov operator has acted on probability distributions by being applied on the left such that

$$(\psi M)(y) = \sum_{x \in S} \psi(x) p(x, y).$$

The same transition kernel also induces an operator on functions. Represent a function $h : S \rightarrow \mathbb{R}$ as the column vector

$$h = \begin{bmatrix} h(x_1) \\ \vdots \\ h(x_N) \end{bmatrix}.$$

Then the matrix multiplication on this column vector gives

$$(Ph)(x_i) = \sum_{j=1}^N p(x_i, x_j) h(x_j).$$

Related to this, we use a rightward Markov operator that is equivalent to a matrix multiplication on a column vector. This form of representation is particularly useful because it gives a direct connection between the Markov operator and conditional expectations.

Definition 8. For any function $h : S \rightarrow \mathbb{R}$, define the action of the Markov operator on h by

$$Mh(x) = \sum_{y \in S} p(x, y) h(y) \quad \forall x \in S.$$

Recall that if a random variable Y has distribution ϕ , then

$$\mathbb{E}[h(Y)] = \sum_{y \in S} \phi(y) h(y).$$

For a fixed current state $X_t = x$, the function $p(x, \cdot)$ is the conditional distribution of X_{t+1} . Therefore, the rightward Markov operation

$$Mh(x) = \sum_{y \in S} p(x, y) h(y) \quad \forall x \in S$$

can be interpreted as

$$\mathbb{E}[h(X_{t+1}) | X_t = x].$$

Thus, while the action of M on distributions propagates probabilities forward in time, the action of M on functions gives the expected future value of a function of the state.

Exercise 2. Prove that $M^t h(x) = \sum_{y \in S} p^t(x, y) h(y)$ and argue that this can be interpreted as $\mathbb{E}[h(X_t) | X_0 = x]$.

1.3.4 Duality Between Distributions and Functions

The two actions of the Markov operator are not separate constructions. In the finite-state case, they correspond to the two matrix products. Let $\psi \in \mathcal{P}(S)$ and let $h : S \rightarrow \mathbb{R}$. Then

$$\begin{aligned}
(\psi M^k)h &= \sum_{y \in S} (\psi M^k)(y)h(y) \\
&= \sum_{y \in S} \left[\sum_{x \in S} \psi(x)p^k(x, y)h(y) \right] \\
&= \sum_{y \in S} \sum_{x \in S} \psi(x)p^k(x, y)h(y) \\
&= \sum_{x \in S} \psi(x) \left[\sum_{y \in S} p^k(x, y)h(y) \right] \\
&= \sum_{x \in S} \psi(x)(M^k h)(x) \\
&= \psi(M^k h).
\end{aligned}$$

If $X_0 \sim \psi_0$, then we have

$$\psi_0 M^k h = \mathbb{E}[h(X_k)].$$

More generally, since $\psi_t = \psi_0 M^t$, we have

$$\psi_t M^k h = \psi_0 M^{t+k} h = \mathbb{E}[h(X_{t+k}) | X_t \sim \psi_t].$$

1.4 Joint Distributions

Now consider the **joint distribution** of a Markov process X_t . Consider (X_0, X_1) . Then the joint probability of a sequence can be defined as

$$\begin{aligned}
q_2(x^0, x^1) &= \mathbb{P}\{X_0 = x^0, X_1 = x^1\} = \mathbb{P}\{X_0 = x^0\} \cap \mathbb{P}\{X_1 = x^1\} \\
&= \mathbb{P}\{X_1 = x^1 | X_0 = x^0\} \cdot \mathbb{P}\{X_0 = x^0\}
\end{aligned}$$

by the Law of conditional probability. More formally, this is

$$q_2(x^0, x^1) = \psi(x_0) \cdot p(x^0, x^1)$$

Iteratively applying a similar extension of the sequence leads to

$$q_{T+1}(x^0, \dots, x^T) = \psi(x^0) \prod_{t=1}^{T-1} p(x^t, x^{t+1})$$

That is, the joint probability is simply the repeated application of the kernel to the initial marginal distribution.

As T increases, the state space will be exponentially expanded, which is referred as the *curse of dimensionality*. Using q_{T+1} , we can express the probability of more complex events. Suppose $B \equiv \{(x^0, \dots, x^T) \in S^{T+1} : x^t \leq x^{t+1} \forall t \leq T-1\}$. Then

$$\mathbb{P}B = \sum_{\hat{x} \in B} q_{T+1}(\hat{x})$$

One important special case happens for the events in a tunnel, defined as follows.

Definition 9. Suppose $D^t \subset S$ for all t . A **tunnel** is the subset of the state space such that

$$D^0 \times \dots \times D^T = \{(x^0, \dots, x^T) \in S^{T+1} : x^t \in D^t \forall t\}$$

Lemma 2. The probability of all events in the tunnel is given by

$$\mathbb{P} \cap_{t \leq T} \{X_t \in D^t\} = \sum_{x^0 \in D^0} \psi(x^0) \sum_{x^1 \in D^1} p(x^0, x^1) \dots \sum_{x^T \in D^T} p(x^{T-1}, x^T)$$

Remark 5. The tunnel probability reflects the **Markov property** of the process. In particular, the probability of a path can be computed recursively using only one-step transitions. This means that at each period t , the transition to x^t only depends on the previous state x^{t-1} and not on the full past history. For example, at x_2 , all one must consider is $p(x_1, x_2)$, i.e. one can only consider x_1 and not x_0 .

1.5 Stationary Distributions

In many cases, a Markov chain converges to a distribution over time, often independently of the initial condition. This suggests a notion of stability analogous to global stability in deterministic dynamical systems. The natural question is whether we can formalize this idea in a similar way.

To do so, we shift our perspective. Instead of tracking individual states, we study the evolution of distributions over states. This leads us to consider the dynamical system $(\mathcal{P}(S), M)$, where $\mathcal{P}(S)$ is the set of probability distributions on S , and M is the Markov operator governing their evolution.

Recall that a deterministic dynamical system is given by a pair (S, h) where S is a metric space (S, d) and $h : S \rightarrow S$ is a self-map. In the stochastic setting, the Markov operator $M : \mathcal{P}(S) \rightarrow \mathcal{P}(S)$ plays an analogous role of h by mapping one distribution into another such that

$$\psi M(y) = \sum_{x \in S} p(x, y) \psi(x) \forall y \in S$$

To treat $(\mathcal{P}(S), M)$ as a dynamical system, we must ensure that the space $\mathcal{P}(S)$ has a suitable structure and that the operator M behaves well on this space.

First, note that $\mathcal{P}(S)$ is compact and convex, thus is a well-behaved state space for the dynamics.

Proof. $\mathcal{P}(S)$ is a simplex Δ , which is closed and bounded, thus compact. Also, a simplex Δ is convex. $\square$

Next, we equip $\mathcal{P}(S)$ with a metric. A natural choice is the p -norm with $p = 1$, which leads to the L_1 -distance as follows:

$$\begin{aligned} \|\psi\|_1 &= \sum_{x \in S} |\psi(x)| \quad \forall \psi \in \Delta^{N-1} \\ \Rightarrow d_1(\psi, \psi') &= \|\psi - \psi'\|_1 \end{aligned}$$

Under this metric, the Markov operator M has an important stability property such that it is nonexpansive.

Property 1. *Let M be a Markov operator determined by an arbitrary stochastic kernel p . M is d_1 -nonexpansive on $\mathcal{P}(S)$ such that $\forall \psi, \psi' \in \mathcal{P}(S)$ we have*

$$d_1(\psi M, \psi' M) \leq d_1(\psi, \psi')$$

Proof. □

This property suggest that the dynamics may contract differences between distributions over time, providing a foundation for convergence.

We now turn to the central object of interest. Fixed points of the dynamical system $(\mathcal{P}(S), M)$ are called stationary distributions.

Definition 10. *A distribution $\psi^* \in \mathcal{P}(S)$ is called **stationary** or **invariant** for M if*

$$\psi^* M = \psi^*$$

That is, if it is a fixed point of the dynamical system $(\mathcal{P}(S), M)$

In this sense, a stationary distribution is the stochastic analogue of a fixed point in deterministic systems.

A fundamental result is as follows.

Theorem 1. *Every Markov operator defined over a finite state space has at least one stationary distribution.*

Proof. Note that $h : \psi \rightarrow \psi M$ is d_1 -nonexpansive, thus d_1 -continuous with a compact domain $\mathcal{P}(S)$. Then by Brouwer's fixed point theorem, the dynamical system $(\mathcal{P}(S), M)$ has at least one fixed point, i.e. a stationary distribution. □

Note that there are three ways of computing the stationary distribution of a Markov chain.

1.5.1 Brute Force

The brute force method computes the stationary distribution by repeatedly applying the transition matrix to an initial distribution. That is, starting from an initial distribution ψ_0 , we iterate

$$\psi'_{t+1} = \psi'_t P$$

for a large number of periods. If the Markov chain converges, this sequence approaches the stationary distribution.

One caveat is that this method only works when the chain exhibits convergence from arbitrary initial conditions, a property that will be formalized later as a notion of global stability.

1.5.2 Eigendecomposition

The eigendecomposition method provides another way to compute stationary distributions by exploiting the linear structure of Markov operators. Note that Markov operators are linear operators since

$$\begin{aligned}\psi_{t+1} &= \psi M \\ \Rightarrow \psi'_{t+1} &= \psi'_t P \\ \Rightarrow \psi_{t+1} &= P' \psi_t\end{aligned}$$

Hence, we can analyze it using tools from linear algebra, in particular eigenvalues and eigenvectors.

In deterministic linear dynamical systems, we studied eigenvalues and eigenvectors primarily to understand convergence to the origin. Since the origin is automatically a fixed point of a linear system, the main question was not how to find the fixed point, but whether trajectories converge to it. This convergence is governed by the eigenvalues, i.e. if $|\lambda_i| < 1$ for all i , then the corresponding components decay to the fixed point zero over time. From a different perspective, a nonzero fixed point of a linear operator must satisfy $Av = 1 \cdot v$ and is therefore an eigenvector associated with $\lambda_i = 1$.

The same logic applies to Markov chains, but the state space is now the simplex of probability distributions rather than $\mathbb{R}^N$, so the zero vector cannot be a feasible fixed point. A stationary distribution ψ^* must instead satisfy $\psi^* = P' \psi^*$, and therefore corresponds to the eigenvector of P' associated with the eigenvalue $\lambda_i = 1$. Thus, the eigenvalue $\lambda_i = 1$ identifies the stationary component of the dynamics, i.e. the stationary distribution.

Theorem 2. *A stationary distribution ψ^* satisfies*

$$\psi^* = P' \psi^*$$

which implies that ψ^ is a eigenvector of the transition matrix P' associated with $\lambda_i = 1$. Thus, a Markov chain has as many stationary distributions as the number of eigenvectors associated with*

$\lambda_i = 1$.

Since eigenvectors are defined up to scale, we must normalize the eigenvector so that its components sum up to one, representing a probability distribution. In particular, if v_i is the eigenvector associated with eigenvalue $\lambda_i = 1$, then the stationary distribution is obtained by scaling v_i so that its components sum to one:

$$\psi^* = \frac{v_i}{\sum_j (v_i)_j}$$

Remark 6. *The eigendecomposition method proceeds as follows.*

1. *Eigendecompose the transition matrix P' implied by M*
2. *Scale the eigenvectors associated with $\lambda_i = 1$.*
3. *These scaled eigenvectors are the stationary distribution.*

An advantage of this method is that it can identify stationary distributions even when the chain does not converge from all initial conditions.

1.5.3 Matrix Inversion

The last method to compute stationary distributions is the matrix inversion method. One begins by solving

$$(I_N - P')\psi^* = 0$$

This can be done as follows:

$$\begin{aligned} &\Rightarrow (I_N - P')\psi^* + 1_N = 1_N \\ &\Rightarrow (I_N - P')\psi^* + 1_N\psi^* = 1_N \cdots \text{since } 1_N\psi^* = 1_N \\ &\Rightarrow (I_N - P' + 1_N)\psi^* = 1_N \\ &\Rightarrow \psi^* = (I_N - P' + 1_N)^{-1}1_n \end{aligned}$$

where the inverse in the last equality is guaranteed, since if the matrix were not invertible, the system would fail to admit a fixed point, which would contradict Brouwer's fixed point theorem.

An advantage of this method is that it can be applied even when the chain does not converge from all initial conditions, and there is no need to scale to get the stationary distribution.

1.6 Dobrushin Coefficient and Global Stability

Now that we have characterized a stationary distribution, i.e. a fixed point of the Markov operator, we turn to the question of whether the marginal distribution converges to this stationary distribution and how this convergence relates to global stability.

Definition 11. *The Markov chain $(\mathcal{P}(S), M)$ is **globally stable** if*

1. it has a unique stationary distribution $\psi^* \in \mathcal{P}(S)$
2. $d_1(\psi M^t, \psi^*) \equiv \|\psi M^t - \psi^*\|_1 \rightarrow 0$ as $t \rightarrow \infty$ for all $\psi \in \mathcal{P}(S)$

The natural question is how to verify whether a given Markov chain satisfies global stability. In deterministic dynamical systems, convergence was characterized by studying whether repeated applications of the mapping lead to a fixed point. That is, if

$$\lim_{n \rightarrow \infty} A^n x_t = 0 = x^*$$

then convergence to the fixed point is achieved. In this context, it is useful to consider contractionary mappings, as they guarantee uniformly shrinking distances.

However, the Markov operator M is not a contraction in general. As shown earlier, it is nonexpansive, meaning that it at least does not diverge, but this alone is not sufficient to guarantee convergence. Therefore, we seek additional conditions under which the operator exhibits a form of contraction.

Intuitively, global stability requires that the effect of the initial distribution vanishes over time. That is, the long-run behavior of the system should not depend on where it started. Formally, for $\psi_t(x) = \mathbb{P}(X_t = x | X_0 = \psi_0)$, we expect that

$$\mathbb{P}(X_t = x) \approx \mathbb{P}(X_t = x | X_0 = \psi_0) \forall \psi_0$$

as $t \rightarrow \infty$. That is, the distribution of X_t becomes approximately independent of the initial distribution ψ_0 as $t \rightarrow \infty$.

To quantify this idea, we introduce a measure of how strongly a kernel mixes different initial states, which leads to the notion of the *Dobrushin coefficient*.

Definition 12. The *Dobrushin coefficient* of a stochastic kernel p is given by

$$\alpha(p) = \min \left\{ \sum_{y \in S} \min \{ p(x, y), p(x', y) \} \mid (x, x') \in S \times S \right\}$$

Remark 7. In words, the Dobrushin coefficient is

1. For every pair of states today (x), find the smaller probability of moving into any state tomorrow (y).
2. Sum these probabilities across all states tomorrow, i.e. $\forall y$
3. The Dobrushin coefficient is the smallest of these sums for all pairs of states today.

Example 1. Let

$$P_H = \begin{bmatrix} 0.9 & 0.1 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.5 & 0.5 \end{bmatrix}$$

1. 1st and 2nd row

$$0.1 + 0.1 + 0 = 0.2 = \alpha_{1,2}$$

2. 1st and 3rd row

$$0 + 0.1 + 0 = 0.1 = \alpha_{1,3}$$

3. 2nd and 3rd row

$$0 + 0.5 + 0.1 = 0.6 = \alpha_{2,3}$$

Therefore,

$$\alpha(P_H) = \alpha_{1,3} = 0.1$$

Property 2. 1. $\alpha(p)$ is large when $p(x, dy)$ are similar across x , i.e. if $p(x, y) \approx p(x', y)$

2. $\alpha(p)$ is large when the current state has little influence on future states. For example, if

$$P = \begin{bmatrix} 0.6 & 0.4 \\ 0.5 & 0.5 \end{bmatrix}$$

3. $\alpha(p) = 0$ if there exists $x, x' \in X$ such that $p(x, dy)$ and $p(x', dy)$ have disjoint supports. For example, if

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

4. $\alpha(p) \in [0, 1]$

Proof. We prove the last property. □

Lemma 3. $\alpha(p) = 1$ if and only if $p(x, dy)$ is equal to a constant distribution $q \in \mathcal{P}(S)$ for all $x \in S$.

Proof. □

Lemma 4. If $\alpha(p) = 1$ then $\{X_t\}$ is i.i.d.

Proof. □

Lemma 5. If X_t is i.i.d., it has one stationary distribution such that $\psi^* = q$.

Proof. □

Theorem 3. If p is a stochastic kernel on S with a Markov operator M , then

$$\|\phi M - \psi M\|_1 \leq (1 - \alpha(p)) \|\phi - \psi\|_1 \quad \forall \phi, \psi \in \mathcal{P}(S)$$

Moreover, this bound is the best available in the sense that if $\lambda < 1 - \alpha(p)$, then there exists $\phi, \psi \in \mathcal{P}(S)$ such that $\|\phi M - \psi M\|_1 > \lambda \|\phi - \psi\|_1$.

Remark 8. The first statement of the theorem above implies that if $\alpha(p) > 0$ then M is uniformly contracting with modulus $1 - \alpha(p)$. Since $(\mathcal{P}(S), M)$ is complete, Banach's fixed point theorem implies the global stability of $(\mathcal{P}(S), M)$.

The second statement of the theorem above implies that this rate $1 - \alpha(p)$ is the best available, i.e. the Dobrushin coefficient provides a precise measure of the contraction rate.

Note that if $\alpha(p) > 0$, then $(\mathcal{P}(S), M)$ being globally stable by Banach's fixed point theorem was an important result. In fact, we can say more. The following is our main stability result for finite Markov chains, which clarifies the relationship between the Dobrushin coefficient and global stability.

Theorem 4. (Main Theorem of Global Stability)

Let p be a stochastic kernel on S with Markov operator M . The following statements are equivalent.

1. The stochastic dynamical system $(\mathcal{P}(S), M)$ is globally stable.
2. There exists t such that $\alpha(p^t) > 0$

Now the following are the summary of stability results in a stochastic dynamical system.

Remark 9. • If $\alpha(p) > 0$, we have a uniform contraction with $1 - \alpha(p)$ as the rate of convergence, i.e. $(\mathcal{P}(S), M)$ is globally stable.

- If $\alpha(p) = 0$, then M is not a contraction so that global stability of $(\mathcal{P}(S), M)$ is not guaranteed.
- However, even if $\alpha(p) = 0$, we can still have a globally stable dynamical system. This occurs when the kernel becomes mixing after a finite number of iterations, so that $\alpha^t(p) > 0$ even though initially $\alpha(p) = 0$.

This raises the question of how to characterize systems that fail to exhibit global stability.

Definition 13. Two states x, y are said to **communicate** if there exists t, τ such that

$$\mathbb{P}(X_t = y | X_0 = x) > 0 \text{ and } \mathbb{P}(X_\tau = x | X_0 = y) > 0$$

In words, if there is some probability to move from x to y over some period of time t and there is some probability to move from y to x over time τ .

Example 2. Suppose

$$P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

Then it is possible to go from A to B and also from B to A . Thus, all states communicate.

Definition 14. A Markov chain is called **irreducible** if all states communicate. It is also called **reducible** if some states never communicate.

Example 3. Suppose

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 0.5 & 0.5 \end{bmatrix}$$

Note that

$$P^2 = \begin{bmatrix} 0.5 & 0.25 & 0.25 \\ 0.25 & 0.5 & 0.25 \\ 0.25 & 0.25 & 0.5 \end{bmatrix}$$

Then since every state can eventually reach every other state, the system is irreducible.

Example 4. Suppose

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0.5 & 0.5 \end{bmatrix}$$

Then state 1 is isolated. Also, even though state 2 and 3 are communicative, they do not with state 1. Therefore, the system is reducible.

Definition 15. A Markov chain is called **aperiodic** if it does not cycle predictably. It is also called **periodic** if certain states guarantees moving to other states.

Example 5. Suppose

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Then we have $A \rightarrow B \rightarrow A \rightarrow B \rightarrow \dots$. Thus, since there is a clear cycling behavior, the system is periodic.

Theorem 5. If $(\mathcal{P}(S), M)$ is not globally stable, then it is either reducible or periodic.

Corollary 1. If a Markov chain is irreducible and aperiodic, then $(\mathcal{P}(S), M)$ is globally stable.